DATA COMMUNICATION AND NETWORKS FALL 2025

SEMESTER PROJECT / COMPLEX ENGINEERING PROBLEM (CEP) REPORT
Project Title: Analysis and Optimization of Data Communication Networks (DCN)

Note: Answering guidelines are included with each question. Students are free to supplement these with any other valid techniques, diagrams, tools, or references that strengthen their report.

1. Group Information

Role	Name	Roll No.
Group Leader	Mohammad Ahtesam	23i – 6060
Member 1	Muhammad Ibrahim	23i – 6090
Member 2	Muzammil Ali	23i – 6163
Member 3	Bilal Ahmed Janjua	23i – 6089
Member 4	Muhammad Abdullah	23i – 6118
Member 5	Uneeb Ashraf	23i – 6042
Member 6	Abdul Mohymin	23i – 6057
Member 7	Wajeh Jillani	23i – 6036

Company Name & Address:
5G Innovation Lab, National Science and Technology Park (NSTP), National University of Sciences and Technology (NUST), H-12 Sector, Islamabad, Pakistan

2. Background / Introduction

(Provide a concise overview of the company, its operations, and the role of its DCN.)

The 5G Innovation Lab at National Science and Technology Park (NSTP), NUST is a research and development facility focused on next-generation wireless communication technologies. The lab works on designing, testing, and validating advanced networking solutions, with a strong emphasis on 5G architecture, high-speed data transmission, low-latency communication, and network reliability.

In the context of Data Communication and Networks (DCN), the lab plays a key role in developing and analysing modern network infrastructures. It provides a practical environment for understanding core networking concepts such as routing, switching, network segmentation, traffic management, and wireless communication systems. The facility supports experimentation with real-time data flow, packet transmission, and performance optimization, making it a valuable platform for applied learning and research in modern communication networks.

3. Objectives of the Project

1. Investigate the design, functionality, and performance of the company's DCN.
2. Identify and address network inefficiencies, security vulnerabilities, and performance bottlenecks.
3. Propose innovative solutions justified with research-based evidence and tools.
4. Develop professional documentation to communicate findings and recommendations effectively.

4. Scope of the Project

1. Analyze end-to-end connectivity, including nodes, links, and protocols.
2. Evaluate the company's equipment, vendor choices, and integration challenges.
3. Examine DNS, DHCP, and AAA mechanisms, and identify areas for improvement.
4. Assess IP schemes, routing protocols, and security measures.
5. Propose and document technical solutions backed by data and industry standards.

5. Deliverables

1. A detailed project report covering all aspects of the problem.
2. Supporting diagrams, pictures, and outputs from analysis tools.
3. Recommendations for network optimization and security enhancement.

6. Company Contact / Reference

Name	Designation	Contact Info
M Rehan Mehmood	Accounts Officer (PPS - 07), NUST Chip Design Centre	mrehanmehmood@icloud.com

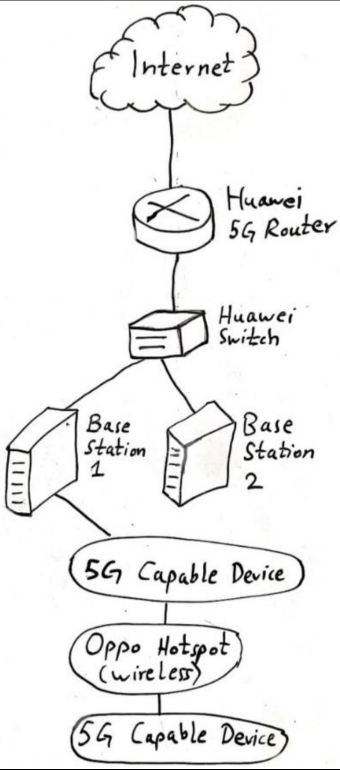
7. Visit Details

Date of Visit	Location / Department	Activities Observed	Photos / Equipment Observed
24/11/2025	5G Innovation Lab, NSTP NUST	Observation of DCN setup, 5G network demonstrations, basic routing and switching concepts, and overview of network monitoring tools	Routers, switches, 5G antennas, network analyzers, photographs of lab equipment and network setup

8. Project Analysis / Questions

Q1. End-to-End DCN Diagram

Question: Draw an end-to-end connectivity diagram of the company’s DCN, including nodes, links, protocols, and their RFCs.

Item	Details
Diagram	
Nodes	Huawei 5G Router, Huawei Switch, 5G Base Stations (2 units), 5G Capable Devices, Oppo Hotspot
Links	• **Router to Switch:** Wired Ethernet • **Switch to Base Stations:** Wired Ethernet • **Base Stations to 5G Capable Device:** 5G Wireless • **Hotspot to 5G Capable Device:** 5G Wireless
Protocols / RFCs	• **TCP/IP:** Internet communication • **DHCP:** Dynamic IP allocation (RFC 2131) • **DNS:** Domain name resolution (RFC 1035) • **HTTP/HTTPS:** Web communication (RFC 2616) • **SNMP:** Network monitoring (RFC 1157)

Q2. Equipment and Vendors

Question: List equipment used, vendors, specifications, and include pictures.

Equipment	Vendor	Specifications	Picture
Huawei 5G Router	Huawei	5G NR support, Gigabit Ethernet ports, Wi-Fi 6, Supports TCP/IP, DHCP, DNS	
Huawei Switch	Huawei	24-port Gigabit Ethernet, Managed Switch, VLAN support, SNMP	
5G Base Station (×2)	Huawei	5G NR, Sub-6 GHz & mm Wave, Supports multiple 5G devices	
Oppo Hotspot Device	Oppo	Portable 5G hotspot, LTE/5G support, Wi-Fi 5, Battery-powered	

Equipment	Vendor	Specifications	Picture
5G Capable Device	Various (Lab)	Smartphone/Laptop with 5G connectivity, Wi-Fi 5/6, TCP/IP compatible	Not Available due to Security Policy

Q3. Applications and Services Supported

Question: Identify applications and services supported, with relevant protocols and RFCs.

Application / Service	Protocol / RFC	Devices Involved
Web Browsing / HTTP Access	HTTP/HTTPS (RFC 2616)	5G Capable Devices, Huawei 5G Router
Dynamic IP Assignment	DHCP (RFC 2131)	5G Capable Devices, Huawei 5G Router
Domain Name Resolution	DNS (RFC 1035)	5G Capable Devices, Huawei 5G Router
Network Management Monitoring	SNMP (RFC 1157)	Huawei Switch, Base Stations, Router
File Transfer / Downloads	TCP/IP (RFC 793), FTP	5G Capable Devices, Servers
5G Wireless Communication	5G NR (3GPP Standard)	5G Base Stations, 5G Capable Devices, Hotspot
IoT Device Communication	TCP/IP, MQTT (RFC 6147 / 6455)	IoT Devices, 5G Base Stations, Router

Q4. Routing Protocols and IP Schemes

Question: Document routing protocols used and their RFCs.

Protocol	RFC / Standard	Devices / Remarks
TCP/IP	RFC 791 / 793	According to lab information, TCP/IP is the core protocol used for network communication across router, switch, and 5G-capable devices.
DHCP	RFC 2131	Lab staff indicated that DHCP is used to assign IP addresses dynamically to connected devices.
Static IP Scheme	N/A	Base stations and servers typically use fixed IP addresses for stability.
DNS	RFC 1035	DNS resolves domain names to IP addresses; the router may act as a DNS forwarder.

Protocol	RFC / Standard	Devices / Remarks
5G NR / Cellular Routing	3GPP 5G Standard TS 23.501	5G base stations route mobile traffic to core network and 5G-capable devices.
NAT / PAT	RFC 3022	Routers may perform address translation for internal devices to access the Internet.

Q5. DNS, DHCP, and AAA Mechanisms

Question: Explain DNS, DHCP, and AAA mechanisms in the DCN.

Service	Details / Configuration	Devices / Remarks
DNS	DNS resolves domain names into IP addresses for network access.	Huawei 5G Router, 5G-capable devices, servers (as per lab information)
DHCP	DHCP assigns IP addresses, subnet mask, gateway, and DNS server details dynamically.	Huawei 5G Router, 5G-capable devices, IoT devices
AAA	Authentication, authorization, and accounting are part of network security. Some details regarding AAA were not provided due to security restrictions.	5G-capable devices, router/base stations

Q6. Network Security Measures

Question: Discuss network protection and security measures, including protocols and RFCs.

Security Measure	Details / Configuration	Devices / Protocols
Firewall	Firewalls filter network traffic. Staff indicated that the router and switch include basic firewall functionality.	Huawei 5G Router, Huawei Switch; TCP/IP, UDP filtering, RFC 793 (TCP), RFC 768 (UDP)
VPN	VPN provides encrypted remote access. The router may support VPN, but exact configurations were not disclosed.	Huawei 5G Router (potential VPN server), 5G-capable devices (potential clients); IPsec (RFC 4301), SSL/TLS (RFC 5246)
IDS/IPS	IDS/IPS monitors network traffic for threats. Presence and	router/switch; TCP/IP, SNMP (RFC 1157)

Security Measure	Details / Configuration	Devices / Protocols
	configuration were not shared due to security concerns.	

Q7. Performance Analysis

Question: List analysis tools used, weaknesses observed, and propose improvements with proper references.

Tool / Method	Weaknesses Observed	Proposed Improvement / Reference
Ping / Traceroute	Lab staff explained these are used to measure end-to-end latency and packet loss, but measurements were not performed during our visit.	Periodic connectivity tests could be implemented to identify slow or unreliable links and optimize routing or network configuration (RFC 792, RFC 1393).
SNMP Monitoring	Some network monitoring data is restricted due to security policies, so full insights could not be shared.	Centralized SNMP monitoring with proper access can detect bottlenecks and ensure timely maintenance (RFC 1157).
Network Analyzer / Wireshark	Packet-level issues could not be observed; real-time traffic capture is restricted.	Controlled packet capture in a test environment can help detect errors, packet drops, and optimize traffic handling (Comer, *Computer Networks*, 6th Edition).
Throughput / Bandwidth Testing Tools	Actual 5G wireless performance, throughput, and congestion could not be measured during the visit.	Conduct controlled throughput and load testing to identify bottlenecks and adjust bandwidth allocation or QoS settings (3GPP 5G NR performance studies).

Q8. Reliability, Redundancy, and Fault Tolerance

Question: Describe reliability and fault tolerance measures (redundancy, backup links, load balancing).

Aspect	Details
Backup Links	Lab staff explained that critical connections, such as router-to-switch and base station links, may have backup links to maintain

Aspect	Details
	network availability in case of failure. Specific details were restricted.
Load Balancing	We were informed that load balancing can distribute traffic across multiple links or devices to avoid congestion and improve performance. Implementation details were not shared due to security concerns.
Failover Mechanisms	The network uses failover strategies to ensure continuous service in case of device or link failure. Exact configurations or active failover mechanisms were not disclosed during our visit.

Q9. Bandwidth Requirements and Management

Question: Determine bandwidth requirements and management strategies.

Link / Segment	Bandwidth	Utilization	Remarks / Optimization
Router to Switch	High-capacity Ethernet (as explained by staff)	Not measured	Critical backbone link; could be optimized with QoS to prioritize important traffic.
Switch to 5G Base Stations	Fiber or high-speed Ethernet	Not measured	Staff indicated that link capacity must support multiple devices; bandwidth management may include load balancing.
Base Station to 5G Capable Devices	5G Wireless (as per 3GPP 5G NR standards)	Not measured	Wireless bandwidth depends on device density and signal quality; traffic prioritization and scheduling improve performance.
Hotspot to 5G Capable Device	5G Wireless	Not measured	Staff noted that hotspot connections are limited by device capabilities; proper allocation of bandwidth can prevent congestion.

Q10. Quality of Service (QoS) Handling

Question: Explain Quality of Service (QoS) handling for critical applications (VoIP, video conferencing, etc.).

Application	QoS Mechanism	Priority / Remarks
VoIP	Staff explained that traffic can be prioritized using QoS policies to reduce latency and ensure call clarity.	High priority; critical for voice communication.

Application	QoS Mechanism	Priority / Remarks
Video Conferencing	QoS rules allocate sufficient bandwidth and prioritize video packets to minimize lag and jitter.	Medium–high priority; important for smooth video streams.
Real-time Services	Real-time applications (e.g., live monitoring, IoT alerts) can be prioritized to maintain timely data delivery.	High priority; delays may impact operations.

Q11. Monitoring and Management Tools

Question: Document monitoring and management tools (e.g., SNMP, NetFlow, Wireshark) used by the IT team.

Tool	Purpose / Usage	Screenshot / Output
SNMP	Used by the IT team to monitor network devices, track performance, and detect faults.	Not provided; information explained by staff.
NetFlow	Collects traffic flow information for analysing bandwidth usage and identifying network bottlenecks.	Not provided; discussed during visit.
Wireshark	Packet-level analysis tool for troubleshooting and network diagnostics. Staff explained its use in controlled environments.	Not provided; access restricted for security reasons.
Network Management Console	Centralized dashboard for monitoring devices, links, and alerts across the DCN.	Not provided; demonstrated conceptually by staff.

Q12. Scalability and Future Growth Considerations

Question: Evaluate scalability and future growth considerations of the DCN.

Aspect	Current Status	Recommendations
Network Size	The DCN currently supports the existing number of 5G devices, base stations, routers, switches, and IoT devices.	Consider planning for additional devices and users by segmenting the network into VLANs or using hierarchical design to maintain performance.
Equipment Capacity	Routers, switches, and base stations have adequate capacity for current traffic loads.	Future upgrades may include higher-capacity switches, routers, or additional base stations to handle increased traffic.
Future Expansion	Staff explained that expansion is possible, but specific plans were restricted due to security concerns.	Plan for modular expansion of hardware and bandwidth, and ensure proper IP addressing and QoS policies to accommodate growth.

Q13. Disaster Recovery Measures and Backup Strategies

Question: Describe disaster recovery measures and backup strategies.

Backup Type	Frequency	Storage Location	Remarks
Configuration Backup	Periodic backup of router, switch, and base station configurations	Local storage on network management server	Staff explained this is done regularly to restore device configurations in case of failure.
Data Backup	Scheduled backups for servers and critical databases	Centralized server or cloud storage	Details of exact frequency and storage locations were restricted for security reasons.
Redundant Links	Backup links are maintained for critical connections	N/A	Ensures network continuity in case of link or device failure.
Power Backup / UPS	Continuous monitoring of power supply	On-site UPS systems	Staff explained that UPS units provide temporary power during outages to prevent network downtime.

Q14. Cloud Integration or Virtualization

Question: Identify cloud integration or virtualization (if any) in the DCN.

Service / Virtualization	Details	Devices / Remarks
Virtualized Network Functions (VNF)	Staff explained that some network functions, such as routing or monitoring, may be virtualized to improve scalability and manageability.	Likely hosted on servers or network management infrastructure; exact details restricted.
Cloud-based Services	Some data or network services may be managed or backed up via cloud platforms for redundancy and accessibility.	Staff indicated use of cloud storage for backups or monitoring; specific providers or configurations were not disclosed.
Server Virtualization	Virtual machines may be used to run network management tools, monitoring software, or test environments.	Observed/explained on lab servers; exact configurations restricted for security reasons.

9. Proposed Optimizations / Recommendations

Problem / Issue	Proposed Solution	Justification / Reference
Limited redundancy on critical links	Introduce backup links or redundant paths for router-to-switch and switch-to-base station connections.	Enhances network reliability and fault tolerance (Comer, *Computer Networks*, 6th Edition).
Potential congestion on 5G wireless links	Implement QoS policies and traffic prioritization for critical applications (VoIP, video).	Ensures low latency and stable performance for time-sensitive traffic (RFC 4594).
Restricted network monitoring visibility	Use centralized SNMP monitoring with proper access levels to track	Provides comprehensive monitoring while respecting security policies (RFC 1157).

Problem / Issue	Proposed Solution	Justification / Reference
	performance and detect faults.	
Scalability limitations for future growth	Plan modular expansion with additional switches, routers, and base stations; implement VLAN segmentation.	Supports more devices and higher traffic without degrading performance (3GPP 5G NR, standard DCN design practices).
Backup and disaster recovery limitations	Schedule regular configuration and data backups; maintain offsite/cloud storage for critical data.	Ensures faster recovery in case of failures or disasters (ITIL best practices, lab staff explanation).

10. Benefits / Learning Outcomes

Learning Outcome	Details
TCP/IP & OSI Understanding	The staff explained how TCP/IP protocols operate in the DCN, and how devices communicate over different layers. This reinforced theoretical knowledge from coursework.
Addressing & Routing	Learned about dynamic IP allocation (DHCP), static IP usage for stability, and routing of traffic between 5G base stations, routers, and devices, as explained by the staff.
Security & Optimization	Gained insights into network security measures such as firewalls, VPNs, AAA mechanisms, QoS, and potential optimization strategies, even though exact configurations were restricted.
Team Collaboration	Working with group members during the visit encouraged discussion, questioning, and collective understanding of DCN concepts and practices.

11. Conclusion

The visit to the company's DCN provided valuable insights into the design, operation, and management of a modern 5G-enabled network. While hands-on testing was limited due to security restrictions, the staff's explanations allowed us to understand the network's components, protocols, applications, security measures, and fault-tolerance mechanisms. Key learnings include the practical application of TCP/IP, DHCP, DNS, QoS, and monitoring tools in real-world networks, as well as strategies for scalability, backup, and optimization. Overall, the session enhanced our theoretical knowledge, highlighted operational practices, and emphasized the importance of security and reliability in DCN management.

12. Course Learning Outcomes (CLOs)

CLO Number	CLO Description	PLO Mapping
1	**Describe** the basic building block of a typical end-to-end Data Communication Network in the light of TCP/IP and OSI layers.	1
2	**Incorporate** four addressing schemes, error detection, and error correction techniques in different Layers of the TCP/IP Model.	1
4	**Incorporate** Transport Layer and Network Layer Protocols for Application layer protocols.	3

13. Group Contributions

Member Name	Member Roll No.	Class Section	Contribution
Mohammad Ahtesam	23i – 6060	A	Collected information on network devices, protocols, and applications; asked questions about routing, IP schemes, and network connectivity.
Muhammad Ibrahim	23i – 6090	A	Inquired about security measures, QoS handling, and monitoring tools; recorded observations on network management.
Muzammil Ali	23i – 6163	A	Noted details about network performance, reliability, and fault-tolerance; asked questions on backup links and failover mechanisms.
Bilal Ahmed Janjua	23i – 6089	A	Gathered information on disaster recovery, cloud integration, and virtualization; asked questions about scalability and future growth.
Muhammad Abdullah	23i – 6118	A	Focused on applications and services supported; asked questions on DNS, DHCP, and AAA mechanisms; documented responses.
Uneeb Ashraf	23i – 6042	A	Observed network topology, diagram connections, and link types; asked questions on

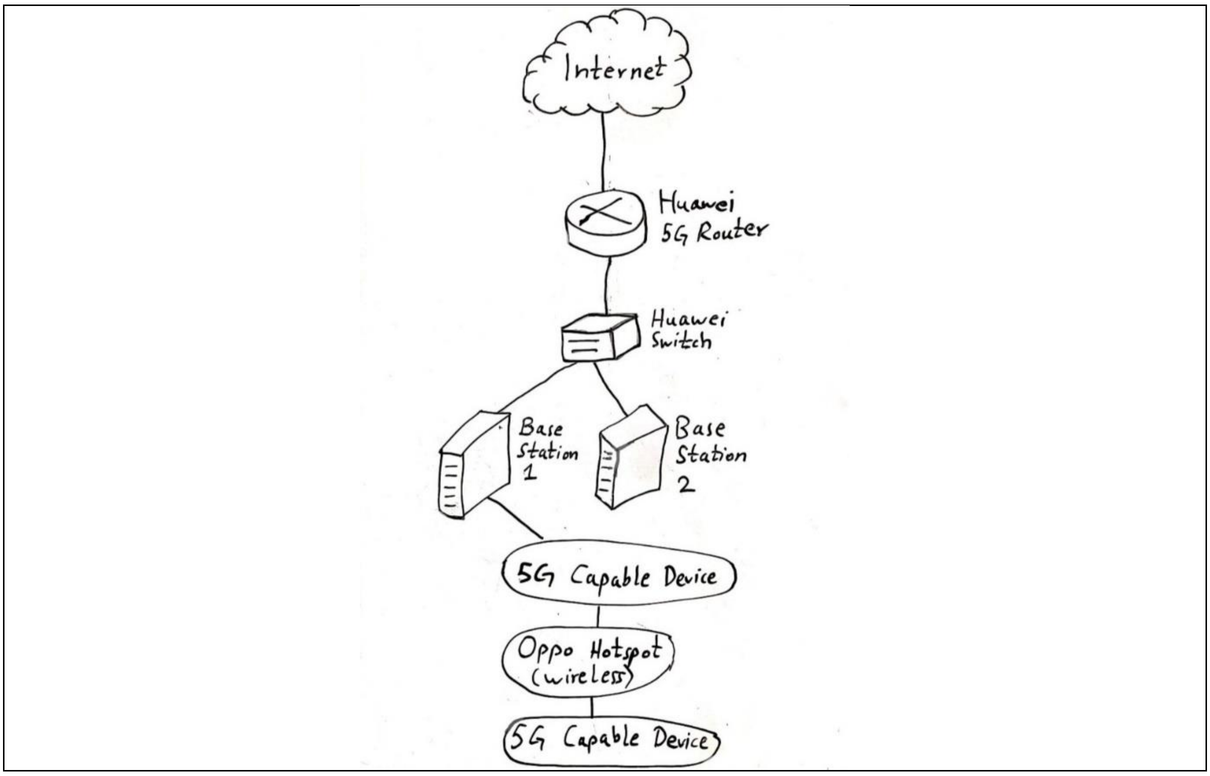
Member Name	Member Roll No.	Class Section	Contribution
			bandwidth and 5G wireless connections.
Abdul Mohymin	23i – 6057	A	Observed the interaction between 5G base stations and core network elements and asked questions about signal handling and handover concepts.
Wajeh Jillani	23i – 6036	A	Took notes on the physical layout of network equipment and discussed practical differences between wired and wireless links.

14. References

1. RFC 791 – *Internet Protocol (IP), DARPA, 1981.*
2. RFC 793 – *Transmission Control Protocol (TCP)*, DARPA, 1981.
3. RFC 768 – *User Datagram Protocol (UDP)*, DARPA, 1980.
4. RFC 2131 – *Dynamic Host Configuration Protocol (DHCP)*, Droms, 1997.
5. RFC 1035 – *Domain Names – Implementation and Specification (DNS)*, Mockapetris, 1987.
6. RFC 2616 – *Hypertext Transfer Protocol – HTTP/1.1*, Fielding et al., 1999.
7. RFC 1157 – *Simple Network Management Protocol (SNMP)*, Case et al., 1990.
8. RFC 3022 – *NAT Terminology and Considerations*, Srisuresh & Egevang, 2000.
9. RFC 4301 – *IPsec Security Architecture*, Kent & Atkinson, 2005.
10. RFC 5246 – *The Transport Layer Security (TLS) Protocol Version 1.2*, Dierks & Rescorla, 2008.
11. RFC 4594 – *Configuration Guidelines for DiffServ Service Classes*, Baker et al., 2006.
12. RFC 6147 – *MQ Telemetry Transport (MQTT) Protocol Specification*, 2011.
13. 3GPP TS 23.501 – *System Architecture for the 5G System*, 3GPP, 2020.
14. Comer, D. E., *Computer Networks and Internets*, 6th Edition, Pearson, 2014.
15. ITIL Foundation, *ITIL Service Management Practices*, AXELOS, 2019.
16. Lab visit and staff explanations, 5G Innovation Lab, National Science and Technology Park (NSTP), (NUST), 24/11/2025.

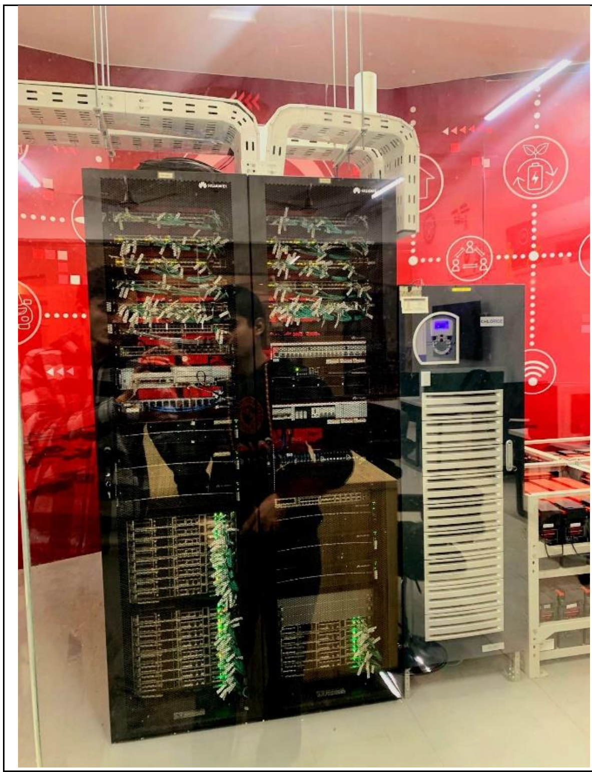
15. Appendices

- Network diagrams



- Visit photographs

- Tool outputs / screenshots



16. Additional Information

Observation / Comment	Details
Security Restrictions	Certain details about network configuration, monitoring tools, and active device settings were not shared due to security policies.
Limited Hands-On Access	Students were not allowed to perform tests, measure bandwidth, or capture live traffic; all information was gathered through staff explanations and observations.
Learning Focus	The visit emphasized understanding network design, protocols, and operational concepts rather than performing practical experiments.
Group Coordination	Team members collaborated by asking questions, taking notes, and discussing observations to compile a comprehensive report.